

MKTG 566

# Digital Marketing Analytics

Spring 2025

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Paccar 483

## Course Information

|                        |                        |
|------------------------|------------------------|
| <b>Class times:</b>    | Wednesday 6pm - 9:20pm |
| <b>Class location:</b> | Paccar 394             |
| <b>Office hours:</b>   | By appointment         |

## Course Objectives

Digital Marketing Analytics uses an analytics-centered approach to improve online advertising, email marketing, and mobile marketing.

This course introduces the core concepts of digital marketing, discusses how these industries operate, and then addresses how companies can make better digital marketing decisions to improve their profitability. Much of the focus is on careful measurement of marketing effectiveness by using data analytics and A/B experiments.

The course is targeted at students considering careers in technology, consulting, and marketing, as well as those who are planning to start their own companies. Students will learn how to improve outcomes for companies that are involved in the ecosystem of digital marketing, whether as a buyer of digital marketing, a seller of digital marketing, an intermediary, or a competitor.

Digital marketing is a data-centric industry in which managers have to rely upon careful

data analysis in order to make decisions. As a result, this course requires hands-on analysis of digital marketing data. While students will employ quantitative methods in the course, the goal is not to produce experts in statistics; rather, students will gain the competency to interact with and manage a marketing analytics team.

The course uses a combination of lectures, cases, and case assignments to learn the material. Students will learn all stages of the analytics process: how to identify marketing problems that can be addressed with analytics, how to decide which analytical method is best suited for a specific marketing problem, how to conduct the analysis using modern-day statistical and machine learning methods in R, how to interpret the results, and how to use the analytical results to make better marketing decisions. This course takes a very hands-on approach with real-world databases and equips students with tools that can be used immediately on the job.

This course focuses specifically on how to measure and improve outcomes when doing online search advertising, online display advertising, mobile advertising, and email advertising. Students who are interested in other marketing analytics topics such as customer acquisition, customer retention, targeted promotions, automated product development, and algorithmic price optimization are encouraged to also take the other classes in the Foster marketing analytics specialization: *Customer Analytics*, *Analytics for Marketing Decisions*, and *Pricing Strategy and Analytics*.

## Course Organization

This course is organized around six modules, each of which are core topics in digital marketing analytics:

**Search advertising:** We discuss how to bid on keywords and brandnames, as well as how to measure the success of a search advertising campaign.

**Display advertising:** We address how the market for display ads works, how companies can buy display ad impressions, and how to measure whether display ads are effective.

**Experiments:** Measuring the effectiveness of an advertising campaign typically requires us to run A/B experiments. We discuss how to set up these experiments and how to analyze their results, as well as what to do in situations where experimentation is not feasible.

**Attribution:** Understanding whether an advertisement *caused* someone to make a purchase is a difficult problem with advertising measurement. We discuss different approaches

to the attribution problem and explain how they can help us better measure the effectiveness of an ad campaign.

**Mobile and email:** We address the challenges and benefits involved with targeting consumers through mobile devices and through email, as well as how to measure the effectiveness of these efforts.

**Privacy and fairness:** We discuss how companies can adapt their digital marketing to maintain standards related to consumer privacy and fairness while also targeting consumers who have an interest in their products.

## Grading Policy and Assignments

The final grade in this course will be based on the following components:

|     |                                            |
|-----|--------------------------------------------|
| 20% | Individual exercises (lowest gets dropped) |
| 40% | Individual homework assignments            |
| 10% | Class participation                        |
| 30% | Final exam (take home)                     |

In all elective courses at Foster, the distribution of final grades must satisfy school-wide requirements. Grades will be curved accordingly at the end of the quarter.

### Assignments:

Much of the learning during the course will happen with the help of homework assignments. For homework assignments, you are allowed to talk to other students; however, you are not allowed to see each other's code, writeup or solutions – the write-up should reflect your own work only.

Fluency in analytics requires both an ability to analyze data and an ability to explain the findings to a managerial audience. The assignments in this course will typically be a combination of both skills: using R to analyze data, and then explaining what actions a company should take. A combination of programming skills, statistical knowledge, and managerial intuition will be required.

We will discuss each homework assignment in class on the day that it is due. As a result, I cannot accept any late assignments.

### Class participationn:

Quality contributions which are relevant to the discussion will improve your participation grade. I will call on students at random to open discussions about assignments and

in-class examples. Your class participation score will be significantly hurt if you are asked to offer your analysis on a case or assignment question and you are not prepared.

**Attendance and punctuality:**

Learning to articulate your analysis and to evaluate and respond to the analysis of others is an important part of what you will learn in this class. If you miss class, you will miss this, and there is not a way to “make it up.” As a result, you should make every effort not to miss class.

If you miss class or are late, this limits your ability to participate and therefore it will lower your class participation grade. If you must miss class, you should do the readings, prepare and turn in the assignments on time (late assignments will not be accepted), and arrange to get notes from a friend about what you missed in class.

## Course Materials

You should prepare for each class session by watching the lecture videos and doing the reading. These materials are specified on Canvas.

Readings in this course consist of a mixture of cases and articles – no textbook is required.

**Textbook:** None required

**Course pack:** <https://hbsp.harvard.edu/import/1282662>

**Lecture notes:** Distributed on Canvas

**Articles and additional course material:** Distributed on Canvas

The syllabus, lecture notes, and additional course material are posted on Canvas. All material is organized by the module it belongs to. Lecture notes will be posted on Canvas on the day of the lecture.

## Software

This course uses the statistics program “R” during the course. R is a very widely used program, and learning how to conduct analysis in R is a major benefit from this course. R is free to use, and has a strong community of users who provide online help and support.

Learning a new programming language can be a challenging task. If you want additional support, I can help with troubleshooting or basic R advice. If you want to learn more about R or want a textbook to use as a reference, I will be happy to provide some suggestions.

## Honor Code

Students are expected to respect the Foster code of conduct at all times.

It will be a violation of academic integrity if you base any of your assignments on discussions or solutions that you find online or that are shared by people who have taken similar courses in the past. I reserve the right to fail you for the course if I become aware of such a violation. To ensure that future classes benefit from a similar learning experience, do not share any case or assignment materials with students outside the class. You are strictly prohibited from sharing content from any assignment with any website, app, or course content repository. This prohibition applies both during the quarter that you are taking this course and any time after the course ends.

## Religious Accommodations

Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at [Religious Accommodations Policy](https://registrar.washington.edu/staffandfaculty/religious-accommodations-policy/) (<https://registrar.washington.edu/staffandfaculty/religious-accommodations-policy/>).

Accommodations must be requested within the first two weeks of this course using the [Religious Accommodations Request form](https://registrar.washington.edu/students/religious-accommodations-request/) (<https://registrar.washington.edu/students/religious-accommodations-request/>)

| <b>Class - Date</b> | <b>Class Topic</b>                                                     | <b>Assignments Due Before Class</b>   |
|---------------------|------------------------------------------------------------------------|---------------------------------------|
| Class 1 — Apr 2     | Course Overview; Display Advertising Concepts and Strategies           |                                       |
| Class 2 — Apr 9     | Display Analytics, Experiments I; Criterion Channel Case               | Analytics Foundation HW; Criterion EX |
| Class 3 — Apr 16    | Experiments II, ATT and ITT Estimates; Uber Eats Case                  | Uber Eats EX; Criterion HW            |
| Class 4 — Apr 23    | Propensity Scores; AdVantage Case                                      | Uber Eats HW; AdVantage EX            |
| Class 5 — Apr 30    | Power Analysis, p-Hacking and MHT; Big Skinny Case                     | Big Skinny EX; p-Hacking EX           |
| Class 6 — May 7     | Search Advertising Concepts and Strategies; Air France Case            | Rocket Fuel HW                        |
| Class 7 — May 14    | Search Advertising Analytics, ID-less Analytics and Privacy; eBay Case | eBay EX; Bazaar EX                    |
| Class 8 — May 21    | Mobile and Email; Artea Case                                           | Starbucks HW                          |
| Class 9 — May 28    | Attribution; Patagonia Case                                            | Patagonia EX                          |
| Class 10 — Jun 4    | Course Wrap-up; Practice Final Exam                                    |                                       |
| <b>Jun 6</b>        | <b>Final Exam</b>                                                      |                                       |

Table 1: Course Schedule and Assignments